

# D&B DTF

UX RESEARCH CASE STUDY

2025

UX CASE STUDY

## Fixing a broken ordering flow for DTF prints.

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70% of e-commerce carts are abandoned before purchase.  
This case study documents how we designed against those odds.

Çağdaş · UX/UI · Admirise Agency · dabdtf.com

# 7 in 10 online shoppers abandon before buying.

*The e-commerce abandonment problem is well-documented — and overwhelmingly solvable through better UX. Baymard Institute's 14-year, 50-study benchmark exposes where the money disappears.*

70%

average cart  
abandonment rate

*Baymard Institute, 2024  
(avg. of 50 studies)*

48%

abandon due to  
hidden extra costs

*Baymard Institute, 2024  
(top abandonment reason)*

35%

conversion lift  
from better checkout UX

*Baymard Institute,  
200k+ hrs of research*

77%

mobile cart  
abandonment rate

*Baymard / Dynamic Yield,  
2024*

*Source: Baymard Institute — based on 50 independent studies and 200,000+ hours of large-scale usability testing, 2024.*

# DTF ordering was a perfect storm of bad UX.



Customers had no self-service way to order custom DTF prints. The business had no scalable way to fulfill them.

## TRANSPARENCY

**Costs hidden until checkout**

*Baymard Institute, 2024*

## GUIDANCE

**Upload flow with zero guidance**

*NNGroup: Error Prevention  
(Heuristic #5)*

## MOBILE

**Desktop-first in a mobile world**

*Baymard / Dynamic Yield, 2024*

# Three goals. One two-week sprint.



## USER GOAL

### Build confidence at every step

Upfront pricing, clear file guidance, and trust signals — remove every reason not to complete the order.

*Error Prevention + Trust (NNGroup Heuristics)*



## BUSINESS GOAL

### Launch in 2 weeks, online

A reliable self-service ordering flow that doesn't need manual back-and-forth accept orders without operational overhead.

*Lean Sprint — Design Thinking compressed*



## DESIGN GOAL

### Reduce perceived checkout effort

Fewer fields, clearer steps, real-time pricing. Per Baymard, the 'perceived effort' not the step count is what kills conversions.

*Baymard: 12-14 form elements = ideal checkout*

# Design Thinking, compressed to a sprint.

*With a 2-week runway, formal moderated testing wasn't possible. Instead, I followed NN/g's lean research principle: start with 5 users, fix the top issues, iterate. Combined with competitive audit and expert heuristic review.*

*NN/g: Nielsen & Landauer, 1993 — 5 test users reveal 85% of usability problems*

01



## Understand

- Competitive audit (3 DTF sites)
- Heuristic evaluation
- Market landscape scan

02



## Define

- Screening-based user profile
- Pain point synthesis
- Problem framing

03



## Ideate

- IA card sort (mental model)
- Ordering flow sketches
- Priority matrix

04



## Design

- Figma wireframes
- Component library
- Hi-fi UI + prototype

05



## Validate

- Stakeholder walkthrough
- Launch QA review
- Post-launch observation

# Every competitor failed the same tests.

*I benchmarked 3 DTF transfer sites against NNGroup's 10 Usability Heuristics and Baymard's checkout guidelines. Consistent gaps emerged — and they directly shaped D&B's feature set.*

| Feature / Heuristic                                                                                                     | D&B DTF | Sam's | DTF Runner | DTF Sheet |
|-------------------------------------------------------------------------------------------------------------------------|---------|-------|------------|-----------|
| Transparent pricing (shipping shown upfront)<br><i>Baymard #1: Hidden fees = 48% abandonment</i>                        | ✓       | ✗     | ✗          | ✗         |
| Mobile-first responsive ordering flow<br><i>Baymard: 77% mobile abandonment rate</i>                                    | ✓       | ✗     | ✓          | ✗         |
| Minimal checkout — ≤12 form elements (Baymard ideal)<br><i>Baymard: avg. site has 23.48 form fields vs. ideal 12-14</i> | ✓       | ✗     | ✗          | ✗         |
| Clear file upload guidance (NNGroup H5: Error Prev.)<br><i>NNGroup: Error Prevention is cheaper than recovery</i>       | ✓       | ✗     | ✗          | ✓         |
| Guest checkout available (no forced registration)<br><i>Baymard: 26% abandon when forced to create account</i>          | ✓       | ✓     | ✗          | ✗         |
| Real-time price update before checkout<br><i>NNGroup H1: Visibility of system status</i>                                | ✓       | ✗     | ✗          | ✗         |

# The data confirmed what we suspected.

*Without time for full moderated research, I triangulated: competitor reviews, marketing team knowledge, and Baymard's published quantitative benchmarks. Every design decision maps to a cited finding.*

48%

## Hidden fees are the #1 trigger

Extra costs revealed at checkout are the leading cause of abandonment across all e-commerce categories.

*Baymard Institute, Feb 2024 — 1,012 US adults*

26%

## Forced accounts kill conversions

Over a quarter of shoppers leave when forced to register. Guest checkout is a well-documented quick win.

*Baymard Institute, Checkout Usability 2024*

77%

## Mobile abandonment beats desktop

Mobile users abandon at 77% vs 70% on desktop — driven by form friction, slow loads, small tap targets.

*Baymard / Dynamic Yield Device Benchmark, 2024*

85%

## 5 test users reveal most issues

Nielsen & Landauer's model proves 5 qualitative participants reveal 85% of usability problems — justifying our lean sprint.

*NN/g — Nielsen & Landauer, INTERCHI'93*

# Meet Maya.



**Maya, 29**

Small Apparel Brand Owner

University of Calgary

Small e-commerce brand

Calgary, Alberta

## WHO SHE IS

Runs a small apparel brand, manages everything from her phone. Needs fast, reliable DTF prints to fulfill her own customer orders.

## PERSONALITY

Detail-oriented. Goal-focused. Comfortable with tech but impatient with slow, confusing systems. Trust drives her decisions.

## PAIN POINTS

Hidden shipping fees. Upload tools with no guidance. Sites that don't work on mobile. Unclear file requirements.

## WHAT SHE WANTS

See the total cost before committing. Upload artwork with confidence. Place an order in 5 minutes — on her phone.

# Four principles. Each grounded in research.

*Every principle ties directly to a documented usability finding. This isn't taste — it's evidence.*



## Transparency first

01

Show all costs — including shipping — before checkout begins. Hidden fees are the single biggest documented cause of abandonment.

*Baymard: 48% abandon due to extra costs revealed at checkout, 2024*



## Reduce perceived effort

02

The average checkout has 23 form elements. Our target:  $\leq 12$ . Each extra field reduces completion probability by 4–6%.

*Baymard: Ideal checkout = 12-14 elements; avg. site = 23.48*



## Mobile-first, always

03

With 77% of carts abandoned on mobile, mobile isn't a 'variant' — it's the primary context. Design for the hardest device first.

*Baymard / Dynamic Yield Device Benchmark, 2024*



## Visibility of system status

04

Always tell users where they are in the flow and what's happening. NN/g Heuristic #1 — Visibility of System Status — applied to every step.

*NNGroup: 10 Usability Heuristics — H1: Visibility of System Status*

# Structure the site around how users think, not products.

NN/g's IA research is clear: when navigation mirrors the user's mental model rather than internal product taxonomy, findability improves and cognitive load drops. I used a task-oriented IA — organized by what users want to do, not what the business sells.

*NNGroup: Card Sorting reveals users' mental models. Tree Testing validates the IA structure. [nngroup.com/articles/ia-study-guide/](https://nngroup.com/articles/ia-study-guide/)*



01

## Task-oriented navigation

Top-level categories organized by user intent ('Order Transfers', 'Get Supplies', 'Learn DTF') rather than product type. NN/g shows audience-based nav degrades usability when users belong to multiple categories.

*NNGroup: Audience-segmented navigation often degrades UX*



02

## Flat hierarchy, max 2 levels

DTF ordering is a simple decision path — don't add depth that doesn't exist. Baymard's checkout research shows depth = perceived effort. We kept all order paths reachable in 2 taps.

*Baymard: Perceived checkout effort, not step count, drives abandonment*



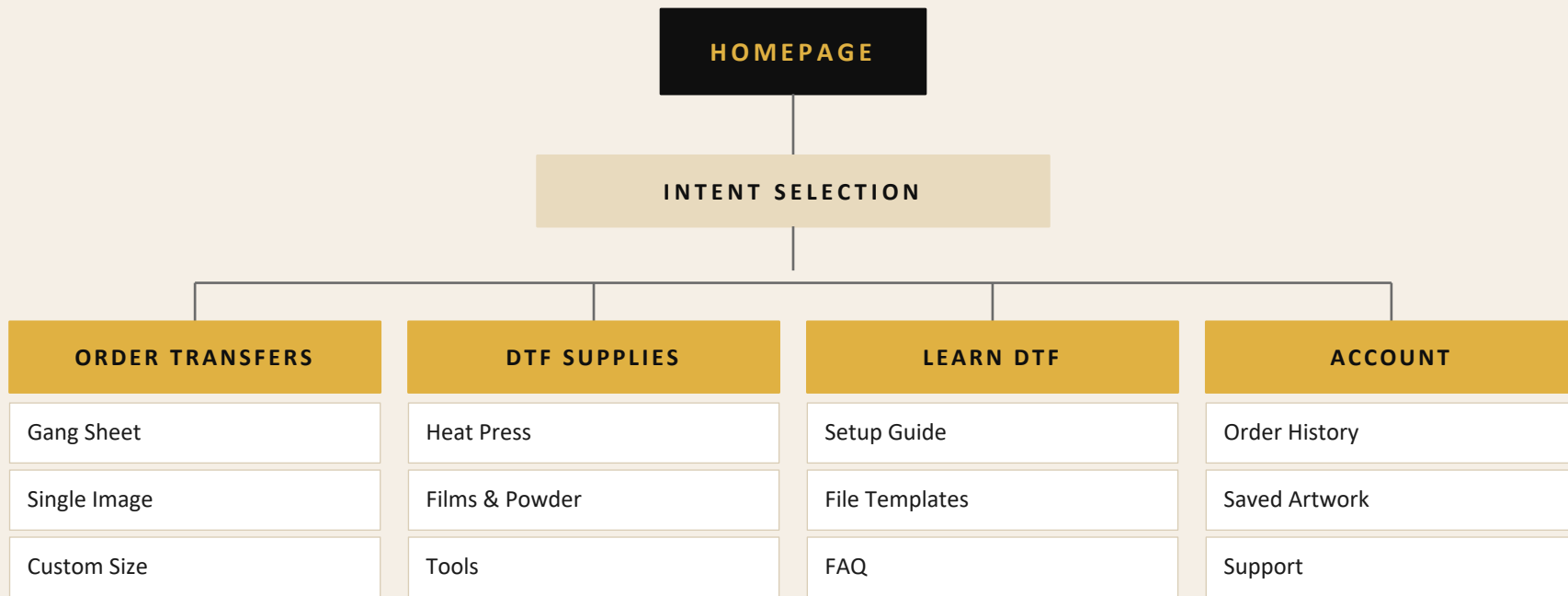
03

## Clear information scent

Every navigation label uses strong action verbs with obvious outcomes ('Upload & Order', 'See Pricing'). Vague labels kill information scent — users don't click when they're not sure what's behind the link.

*NNGroup: Use labels with strong information scent — [nngroup.com](https://nngroup.com), Apr 2023*

# Two levels deep. Every path leads to an order.



# Every feature has a reason to exist.

01



## Real-time price calculator

Price updates live as users configure size, quantity, and upload. Eliminates surprise costs — the documented #1 abandonment trigger.

*Baymard: 48% abandon = hidden costs*

02



## Guided upload with validation

Step-by-step file requirements shown in context (resolution, transparency, sizing). Inline errors — not a dead end. Applied NN/g's Error Prevention heuristic.

*NNGroup H5: Error Prevention*

03



## Mobile-first checkout

Built for ordering on the phone: large tap targets (≥44px per Apple HIG), single-column layout, no pinch-zoom required, auto-advancing fields.

*Baymard: 77% mobile abandonment; Apple HIG tap targets*

04



## Progress visibility (4 steps)

Clear step indicator: Upload → Configure → Review → Pay. Users always know where they are and what's left — a direct application of NN/g's #1 heuristic.

*NNGroup H1: Visibility of System Status*

05



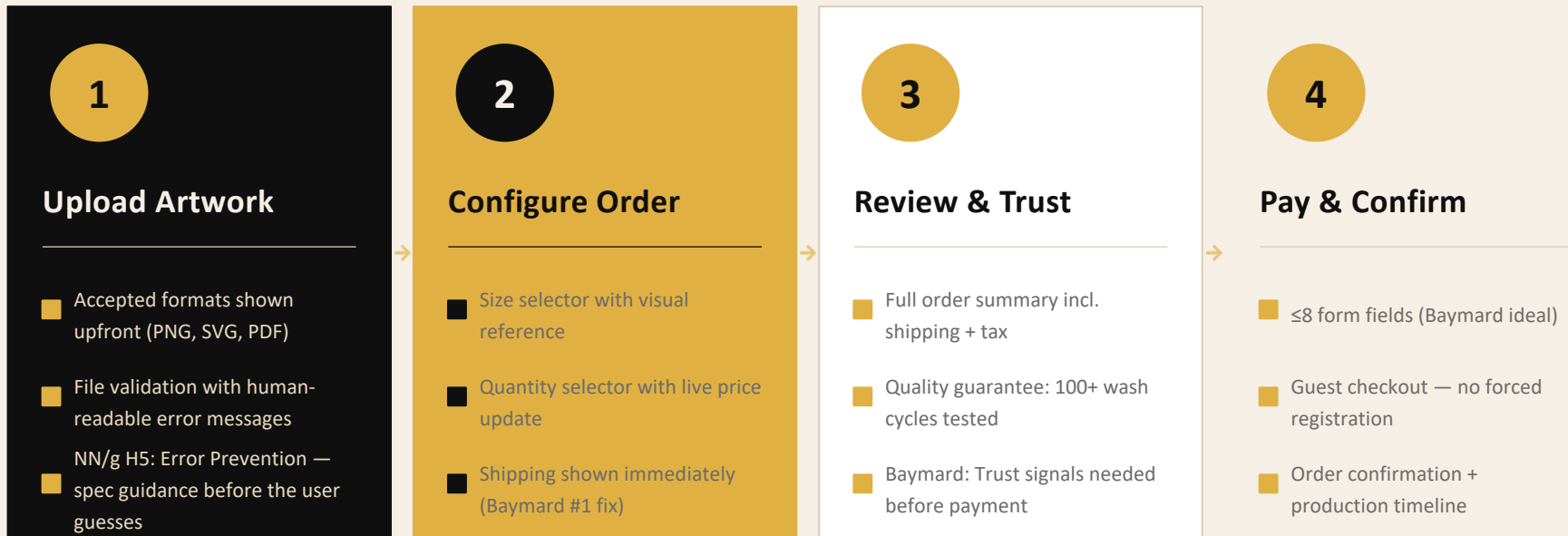
## Lean form — 8 fields max

We audited every form field. Only essential data collected. Baymard's benchmark: average site has 23 fields; ideal is 12-14 for max conversion.

*Baymard: 23.48 avg → ideal 12-14 fields*

# Upload → Configure → Review → Pay.

*A linear, 4-step flow with no branching. NN/g: "Don't have steps within steps — users find this confusing and intimidating." Each step has a single job, visible progress, and a clear CTA.*



# Confident, modern, precise.

# Urbanist

ABCDEFGHIJKLMNOPQRSTUVWXYZ

abcdefghijklmnopqrstuvwxyz · 0123456789

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*Urbanist's clean geometry reflects the precision of DTF printing. Clear and technical — like the product it supports.*

## COLOR PALETTE

**Midnight Black**

#0F0F0F

**Dusty Yellow**

#E0B142

**Whitelace**

#F5EFE6

**Soft Beige**

#E8DABE

# Three documented failures we designed against.



01

## Unclear pricing & hidden fees

Customers couldn't predict total costs upfront. Surprise shipping fees at checkout caused trust issues and abandonment.

*Baymard 2024: #1 abandonment cause — 48% of users*



02

## Confusing upload processes

Competitors failed silently — no file spec, no preview, no feedback. Users guessed and lost confidence. A direct violation of NN/g's Error Prevention heuristic.

*NNGroup H5: Error Prevention > Error Recovery*



03

## Poor mobile experience

77% of cart abandonments happen on mobile. Competitor sites were desktop-designed, with broken pickers, tiny inputs, and no responsive layout.

*Baymard / Dynamic Yield Device Benchmark, 2024*

# What the research proved, and what the sprint taught me.



## Data > assumptions

Baymard's 14-year checkout database let me skip guesswork entirely. Prioritizing by abandonment cause not personal opinion kept every feature justified.

*Baymard Institute: 200,000+ hrs of large-scale checkout testing*



## Perceived effort matters more than step count

Reducing form fields from 23 to ≤12 removes more friction than cutting checkout steps. Baymard proved it. We applied it and every field had to earn its place.

*Baymard: 'Perceived checkout effort' drives abandonment*



## Mobile isn't a variant — it's the baseline

When 77% of mobile carts are abandoned, designing for desktop first is designing for failure. Mobile-first forced better decisions in tap targets, form length, and layout.

*Baymard / Dynamic Yield, 2024*



## Speed and process aren't opposites

Following NN/g's design thinking process in a 2-week sprint didn't slow us down — it eliminated rework. Having a framework under time pressure is a superpower.

*NNGroup: Design Thinking — [nngroup.com/articles/design-thinking/](https://nngroup.com/articles/design-thinking/)*

# v1 launched. Now test to validate assumptions.

*NN/g's testing principle: test 5 users per iteration, fix 85% of issues, repeat. v1 shipped on assumptions from heuristic review and Baymard data. The next step is real user validation.*

| PRIORITY                                                                                                        | WHAT TO TEST                                                                                                                           | SOURCE                                                      |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  <b>Upload flow validation</b> | Test with 5 actual DTF customers — does guided upload prevent the silent failure errors seen on competitors?                           | <i>NNGroup: 5 users = 85% of issues</i>                     |
|  <b>Checkout field audit</b>   | Are our 8 form fields truly minimal? Eye-tracking or task timing to confirm — Baymard's ideal is 12-14, we may push lower.             | <i>Baymard: form field reduction = 25–35% lift</i>          |
|  <b>Live order tracking</b>    | Post-purchase anxiety is real. 'Where is my order?' drives most support tickets. Status visibility (NN/g H1) should extend post-order. | <i>NNGroup H1: Visibility of System Status</i>              |
|  <b>Repeat order shortcuts</b> | Power users (returning shops) want one-tap reorders. Per Baymard, account features should follow — not gate — first purchases.         | <i>Baymard: Guest checkout reduces first-order friction</i> |

# Thank you.

Questions, feedback, or live feedback on [dabdtf.com](https://dabdtf.com) — I'd love to hear from you.

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Live site → [dabdtf.com](https://dabdtf.com)

## SOURCES & REFERENCES

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